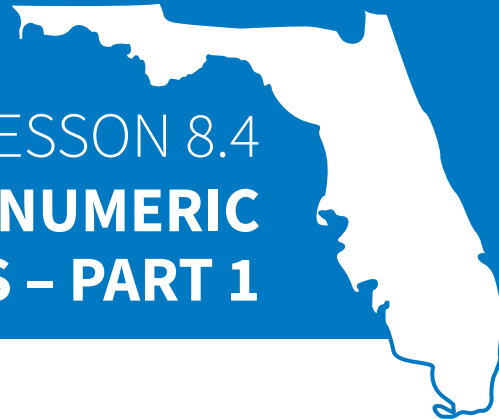




LESSON 8.4

EVALUATING NUMERIC EXPRESSIONS – PART 1

LESSON INTRODUCTION

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

MA.7.NSO.2.2 Add, subtract, multiply and divide rational numbers with procedural fluency.

LEARNING TARGETS

- I can evaluate numeric expressions with up to three operations.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.

MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.

WORKING TOWARD

MA.8.NSO.1.7 Solve multistep mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

ESSENTIAL QUESTIONS

- How can I use order of operations to evaluate numeric expressions?

LESSON NARRATIVE

In this lesson, students extend their knowledge of the order of operations to include what they have learned about exponents and the use of the absolute value symbol. Students begin by getting a refresher on the process of the order of operations that they learned in 6th grade. They will then receive formal instruction on using the order of operations with exponents and grouping with a focus on absolute value notation. Students get the opportunity to practice applying what they have learned by evaluating numeric expressions.

COMPONENT SUMMARY

8.4.1 WARM-UP (~5 MIN)

Math symbols activity

8.4.3 GUIDED INSTRUCTION (15 MIN)

Formal instruction on the order of operations with exponents and absolute value

8.4.5 GUIDED PRACTICE (10 MIN)

Structured practice on using the order of operations and explaining each step

8.4.7 WRAP-UP (~5 MIN)

Cool Down that assesses student understanding of evaluating expressions

8.4.2 REFRESHER (5 MIN)

Refresher on the order of operations

8.4.4 COLLABORATION (10 MIN)

Collaborative practice with evaluating expressions

8.4.6 YOUR TURN! (10 MIN)

Independent practice with using the order of operations to evaluate expressions

RESOURCES

GLOSSARY

Key Terms:

- Absolute value notation

Additional Terms:

- Order of operations

MATERIALS

- (Optional) Calculators
- (Optional) Video

ADDITIONAL PREPARATION

- Consider grouping for 8.4.4 Collaboration.
- The 8.4.1 Warm-Up video is optional

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may forget to complete the calculation inside the absolute value before writing its actual absolute value.
- Students may forget to write the positive value when working with absolute values.
- Students may misunderstand the hierarchy of operations.

THINGS TO CONSIDER

- A horizontal fraction bar implies grouping of the numerator and grouping of the denominator. Students may have difficulty using the order of operations with fractions and division.
- The order of operations is often taught using a mnemonic device as a hard and fast rule. The lesson does not contain any mnemonic devices but instead relies upon both the order of operations and the properties of equality, such as commutative and associative properties. Encourage students' flexible thinking by demonstrating various ways a numeric expression can be evaluated.
- Determine if your students need the support of all or part of the refresher.
- If time is an issue, then consider assigning 8.4.6 Your Turn! for homework.
- Allowing students to use calculators during the lesson may help students focus on discovery and process of evaluating numeric expressions. Consider having no-calculator questions where students turn their calculator over.

LITERACY AND LANGUAGE ROUTINES

Clarify, Critique, Correct

8.4.3 Guided Instruction provides students with formal instruction on evaluating numeric expressions. During this activity, consider providing students with a Think-Aloud so students can see the steps and process being done on a more complex problem.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

8.4.5 Guided Practice provides students with a practice opportunity using the order of operations and properties of equality. Students will use the structure of the numeric expression to decompose a complex problem into manageable chunks in a logical and sequential process.

DIFFERENTIATE INSTRUCTION

For 8.4.4 Collaboration, consider grouping students together based on their learning styles. This will help provide students with a commonality as they work through the questions in the activity. During 8.4.2 Refresher, when students are getting a refresher on the order of operations, consider the use of a foldable or anchor chart. In that, have students notate the steps in the order of operations along with examples. 8.4.7 Wrap-Up will provide students with a Cool Down that assesses their level of understanding with evaluating expressions using the order of operations.

8.4.1 WARM-UP: MATH SYMBOLS

TEACHER MOVES

Consider providing students with several symbols to choose from. Students may not be able to recall symbols that they have seen but do not understand.



8.4.1 Warm-Up: Math Symbols

In the 18th century, Robert Recorde decided to use a symbol for the idea of "is equal to" so he would not have to write the words over and over. As more mathematicians also began to use the symbol, the "=" became accepted as the standard symbol for equality. Math is full of symbols that shorten lengthy words and calculations. But just like symbols you use when communicating with others, like "LOL" in texting, once you learn their meaning, understanding the mathematics is not so challenging.

1. List math symbols that you are familiar with and what they represent.

Answers will vary. $\times$ means multiply, $-$ means subtract, $+$ means add.

2. List other math symbols you have seen before, but you are unfamiliar with.

Answers will vary. $\pm$, $\cup$, $\approx$

**8.4.2 Refresher: Order of Operations**

Math symbols are used to provide instructions on how to apply number relationships and operations. Properties of operations and the order of operations work together to guide the order in which operations are performed.

- Recall that the order of operations is an agreed upon set of rules for how to evaluate an expression. Work with a partner to number each operation so that the order of operations is in the correct order.

- 1 Perform operations inside grouping symbols.
- 3 Perform multiplication and division in order from left to right. ($\rightarrow$)
- 2 Evaluate expressions with exponents.
- 4 Perform addition and subtraction in order from left to right. ($\rightarrow$)

8.4.2 REFRESHER: ORDER OF OPERATIONS**TEACHER MOVES**

This activity is designed to be a brief refresher on the order of operations. Keep in mind that students have worked with the order of operations in prior grade levels. Watch your timing here to avoid being caught up in an extensive review. If students need more understanding, then incorporate that in other places throughout the lesson.

**8.4.3 Guided Instruction: Using Order of Operations**

- Use order of operations to evaluate one of the expressions shown in the table below while your partner evaluates the other expression.

Expression A	Expression B
$(6 + 0.5)^2 \div 0.25$	$(6^2 + 0.5^2) \div 0.25$
$6.5^2 \div 0.25$ $= 42.25 \div 0.25$ $= 169$	$(36 + .25) \div 0.25$ $= 36.25 \div 0.25$ $= 145$

- How are the problems similar?
Answers will vary. The problems are similar by both having exponents and the final step being to divide by 0.25.
- How are the problems different?
Answers will vary. The problems are different in that in expression A, you add the expressions inside the parenthesis before applying the exponent, whereas in expression B you apply the exponents to the expressions inside the parentheses before taking their sum.

Recall that the power of a product exponent law states that when raising a product to a power, distribute the power to each factor.

8.4.3 GUIDED INSTRUCTION: USING ORDER OF OPERATIONS**TEACHER MOVES**

For questions 1 and 2, consider having students discuss them with a partner and then share out their responses in whole group. The responses may be more thorough when provided orally from students than their writing of the explanation in the book.

DIFFERENTIATE INSTRUCTION

Ask students to provide an example of a power of a product exponent law.

8.4.3 GUIDED INSTRUCTION: USING ORDER OF OPERATIONS (CONTINUED)

TEACHER MOVES

Students may need additional strategies for how to approach these types of questions. Consider showing students how to break down each sentence in this series of statements and how to look for correct matches in each one.

ENRICHMENT

How can you determine if two numerical expressions are equivalent?

4. Explain if the power of a product exponent law should have been used to evaluate Expression A.

Answers will vary. No, the power of a product exponent law should not have been used since the values inside the parentheses are being added and not multiplied.

5. Complete the statements about the expressions.

The two expressions are ☐ equivalent
☒ not equivalent

because they ☐ have the same value.
☒ do not have the same value.

For Expression A, first

- ☒ add inside the parentheses, then apply the exponent to the sum.
- ☐ apply the exponent to each addend, then add to find the sum.

For Expression B, first

- ☐ add inside the parentheses, then apply the exponent to the sum.
- ☒ apply the exponent to each addend, then add to find the sum.

For both expressions, addition was done before division

because ☐ the addition is to the left of the division.
☒ the addition is inside the grouping symbols.

TEACHER MOVES

Consider providing additional time to discuss the absolute value component. Some students may have forgotten how absolute value works and need a minor refresher when reviewing.

ENRICHMENT

- What would have happened if the 10 was positive in both?
- Would the answers have been different still? Why or why not?

6. Anna evaluated Expression A and Expression B. Her work is shown.

Expression A	Expression B
$ -10 \div 2 + 10^2$	$-10 \div 2 + 10^2$
$10 \div 2 + 10^2$	$-10 \div 2 + 100$
$10 \div 2 + 100$	$-5 + 100$
$5 + 100$	95
105	

Anna's partner gave her a thumbs down on both of her answers. Why did Anna arrive at two different values when evaluating these similar looking expressions?

In expression A, you have to find the absolute value of -10 , making the value a positive 10.



Absolute value notation should be treated the same as grouping symbols when following the order of operations.



8.4.4 Collaboration: Determining Equivalent Expressions

Answer the questions with your partner.

Tanisha says that when she evaluated an expression, the result was 7.25.

- Use mental math to estimate each value. Then, circle which of the following expressions could have been Tanisha's expression.

$$(6 - 0.5)^2 \times 5$$

$$6 - 0.5^2 \times 5$$

$$6 + (-0.5)^2 \times 5$$

$$5 \times 6 - 0.5^2$$

- Find the values of the expressions to verify your choice.

Expression	Value
$(6 - 0.5)^2 \times 5$	151.25
$6 - 0.5^2 \times 5$	4.75
$6 + (-0.5)^2 \times 5$	7.25
$5 \times 6 - 0.5^2$	29.75

8.4.4 COLLABORATION: DETERMINING EQUIVALENT EXPRESSIONS

TEACHER MOVES

Consider asking students which expressions in question 1 are equivalent. This may give some insight on how much the student has already learned.

TEACHER MOVES

Consider having students turn their calculators over for question 1 and then to use their calculators for question 2. Calculators such as Desmos can determine the value if the student inputs the expression as written. Consider providing students with four-function calculators, where the student will have to use their knowledge to evaluate while the calculator can be a support to learning.



8.4.5 Guided Practice: Using the Order of Operations

- For each table, an expression is evaluated, but the steps are mixed up. Order the expressions by writing the numbers 1-4 next to the expressions shown. Explain the steps in the column on the right. The tables have been started.

4	18.75	Perform Addition
2	$6.75 + 1.5 \times 8$	Evaluate the exponent
1	$6.75 + 1.5 \times 2^3$	Original expression
3	$6.75 + 12$	Perform Multiplication

3	$32.768 - 14.8$	Take the absolute value
1	$ (-3.2)^3 - 14.8$	Original expression
4	17.968	Perform Subtraction
2	$ -32.768 - 14.8$	Evaluate the exponent

- Evaluate each expression.

a. $(-8) \div 2^4 \times 2$

$$(-8) \div 16 \times 2 = -0.5 \times 2 = -1$$

b. $4 \times \left(\frac{3.5}{14}\right)^2$

$$4 \times .25^2 = 4 \times .0625 = .25$$

c. $-5 \times |3.6 + (-4.2)|$

$$-5 \times |-0.6| = -5 \times .6 = -3$$

8.4.5 GUIDED PRACTICE: USING THE ORDER OF OPERATIONS

TEACHER MOVES

For this activity, consider having students remain with their partner from the prior 8.4.4 Collaboration activity. Introduce the question to the class and then release students to complete the question with their partner before going over together as a whole group.

TEACHER MOVES

Consider having students do question 2 without a calculator.

8.4.5 GUIDED PRACTICE: USING THE ORDER OF OPERATIONS (CONTINUED)

TEACHER MOVES

Have students share out their process to completing question 3. Some students will have a misconception with the multiplication and division step. Oftentimes, students will think that multiplication comes before division instead of either/or from left to right. This is a common misconception that comes from teaching tricks and mnemonic devices when teaching the order of operations.

3. Josiah evaluated $a^3 \div 4(6.4 + b)$ when $a = 4$ and $b = -8.4$. His work is shown below.

- a. Circle Josiah's error and choose which step of the order of operations he did not apply correctly.

$4^3 \div 4(6.4 + -8.4)$ $4^3 \div 4(-2)$ $64 \div 4(-2)$ $60 \div (-8)$ -7.5	- Evaluate expressions with exponents. - Perform operations inside grouping symbols. - Perform multiplication and division in order from left to right. - Perform addition and subtraction in order from left to right.
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- b. Discuss your answer with your partner. Then, work together to evaluate the expression correctly.

Answers will vary. Josiah should have divided 64 by 4 before multiplying by -2.

$$\frac{64}{4} = 16(-2) = -32$$

8.4.6 YOUR TURN!

TEACHER MOVES

Consider the use of a Think-Pair-Share strategy with this activity. Allow students the opportunity to try all the questions on their own and then check their work with their partner to see which ones they arrived at the same answer on and which ones they need to discuss further and come to consensus on.

QUESTIONING

For students who finish early, consider having them create expressions and swapping with another classmate.



8.4.6 Your Turn!

1. Evaluate each expression without using a calculator. Write an explanation using the order of operations next to each step.

$ 8 - 10 \div (-12)$	Explanation
$ -2 \div (-12)$	Perform subtraction inside of absolute value symbol
$2 \div (-12)$	Find the absolute value
$-\frac{1}{6}$	Perform division

$(1.5 \times 2)^2 \div 2$	Explanation
$(3)^2 \div 2$	Perform multiplication inside parenthesis
$9 \div 2$	Evaluate the exponent
4.5	Perform division

2. Evaluate each expression.

a. $\frac{(-8 \times 0.25)^4}{5}$

$$\frac{(-2)^4}{5} = \frac{16}{5} = 3.2$$

b. $4 \times |10.6 \times (-0.3)|$

$$4 \times |-3.18| = 4 \times 3.18 = 12.72$$

c. $(-95 \times 0.7)^2 + 8.75$

$$(-66.5)^2 + 8.75 = 4422.25 + 8.75 = 4431$$

3. Determine if the equations in the table are true or false. Show work to support your answer.

Equation	True or False?
$ 7.3 - 8.5 \times (-2.5) = 3$ $-1.2 \times -2.5 = 3$ $1.2 \times -2.5 = 3$ $-3 \neq 3$	False
$2 - 2 \times 2^2 = -6$ $2 - 2 \times 4 = -6$ $2 - 8 = -6$ $-6 = -6$	True
$4.2 \times 3.5 + (-3)^3 = 41.7$ $4.2 \times 3.5 + (-27) = 41.7$ $14.7 + (-27) = 41.7$ $-12.3 \neq 41.7$	False

8.4.6 YOUR TURN! (CONTINUED)

DIFFERENTIATE INSTRUCTION

When reviewing this question as a whole group, have students agree or disagree with true or with false. From there, you will be able to see which students are correct and which are incorrect. Use this information to have students that are correct share their steps to help other students hear their thinking and processes.



8.4.7 Wrap-Up: Cool Down

For each problem, a student was asked to evaluate the expression. Their work is shown in the table. If there are mistakes in the student's work, explain what the mistakes are and write the correct response. If there are no mistakes, leave the rest of the row blank.

Problem	Mistake(s)	Correct Response
1. $7^2 = 14$	Student multiplied the base by the exponent instead of multiplying the base by itself.	49
2. $(8^5)^0 = 8^5$	Student added 5 and 0 instead of multiplying 5 by 0.	1
3. $(-3)^3 = 27$	Student forgot that a negative times a negative times a negative is a negative.	-27
4. $(4 + 5)^2$ $= 4^2 + 5^2$ $= 16 + 25$ $= 41$	Student distributed the exponent in the parenthesis instead of adding the numbers together first.	81
5. $(-4)^5 \cdot (4)^0$ $= (-4)^{5+0}$ $= (-4)^5$ $= -1024$	Student added the exponents but should not have because the bases are not the same.	-1,024
6. $[(8 + 3^2) \times 4]^2$ $= [8 + 9 \times 4]^2$ $= [8 + 36]^2$ $= 44^2$ $= 1936$	The student should have added the 8 and 9 before multiplying by 4 since they are in parentheses.	4,624

8.4.7 WRAP-UP: COOL DOWN

TEACHER MOVES

Students should not use a calculator to complete the Wrap-Up.